

Nonlinear equations and systems

Solving $f(y) = 0$ when no closed formula exists: bisection, Newton, secant and fixed-point iteration — with the error analysis that explains why each one converges as fast as it does.

linear → superlinear → quadratic

7 topic blocks · study reference sheet

01Existence of zeros & conditioning

PROBLEM

Given $f: \mathbb{R} \rightarrow \mathbb{R}$, find y with $f(y) = 0$. Such a y is a root, or zero, of f .

THEOREM — EXISTENCE OF ZEROS

$$f \text{ continuous on } [a,b], \quad f(a) \cdot f(b) < 0$$
$$\Rightarrow \exists y \in (a,b) : f(y) = 0$$

The sign change forces the graph to cross the axis, but the proof says nothing about *where*. That gap is what the numerical methods fill.

CONDITIONING — DERIVATION

Perturb f vertically by ϵ ($\tilde{f} = f + \epsilon$) and let $\tilde{y}$ be the perturbed root, $\Delta x = \tilde{y} - y$. Expanding around y :

$$0 = f(y) + f'(y)\Delta x + O(\Delta x^2) + \epsilon$$
$$f(y) = 0 \Rightarrow \Delta x \approx -\epsilon / f'(y)$$

READING THE ESTIMATE

The smaller $|f'(y)|$ — the flatter the curve at the crossing — the larger the displacement for the same ϵ : the root is **ill conditioned**. Extreme case $f'(y) = 0$ (double root, e.g. $f(x) = x^2$): maximal sensitivity, whatever algorithm is used.

RESIDUAL ≠ ERROR

As with $K(A)$ for linear systems, conditioning is a property of the problem (f, y), not of the algorithm. Since $|f(c_k)| \approx |f'(y)| \cdot |c_k - y|$, a tiny residual is no evidence of a small error when $|f'(y)|$ is small.

02Bisection

HYPOTHESES

f continuous on $[a,b]$ and $f(a) \cdot f(b) < 0$ — exactly the hypotheses of the existence theorem, turned into an algorithm.

CONSTRUCTION

$$c_k = (a_k + b_k) / 2$$
$$f(a_k) \cdot f(c_k) < 0 \Rightarrow a_{k+1} = a_k, \quad b_{k+1} = c_k$$
$$\text{else} \Rightarrow a_{k+1} = c_k, \quad b_{k+1} = b_k$$

Either branch preserves the bracketing and halves the interval — always, regardless of how f behaves inside it.

STOPPING CRITERIA

(a) Residual. $|f(c_k)| < C \cdot \text{Tol}$ — can deceive on ill-conditioned roots: small residual does not imply small error.

(b) Interval width. $|b_k - a_k| < C \cdot \text{Tol}$ — more robust: it never depends on the values of f , only on the geometric count of halvings, and does not even require f to be differentiable.

CONVERGENCE

$$e_k = |y - c_k| \leq b_k - a_k$$
$$b_k - a_k = (1/2)^k (b_0 - a_0)$$

Linear convergence with fixed rate $1/2$, independent of the conditioning of the root.

COST PER DECIMAL DIGIT

Requiring $(1/2)^k \leq 1/10$ gives $k \geq \ln 10 / \ln 2 \approx 3.32$: about 3–4 iterations per digit, always. Slow, but with a guarantee no other method here offers.

03Newton’s method

CONSTRUCTION — TANGENT LINE

$$\text{tangent: } y - f(x_k) = f'(x_k)(x - x_k)$$
$$\text{set } y = 0 \Rightarrow x_{k+1} = x_k - f(x_k) / f'(x_k)$$

STARTING POINT

x_0 can be any point — no bracketing $f(a)f(b) < 0$ required. In exchange convergence is not guaranteed: far from y , or near a point with $f' \approx 0$, the nearly horizontal tangent can throw the iteration arbitrarily far, or make it diverge.

ERROR ANALYSIS — TAYLOR

Expand f around x_k , evaluate at y , second-order Lagrange remainder, ξ_k between x_k and y :

$$0 = f(y) = f(x_k) + f'(x_k)(y - x_k) + \frac{1}{2}f''(\xi_k)(y - x_k)^2$$
$$+ f'(x_k) \Rightarrow y - x_{k+1} = -\frac{1}{2} [f''(\xi_k) / f'(x_k)] (y - x_k)^2$$
$$e_{k+1} \approx M_k e_k^2, \quad M_k = \frac{1}{2} f''(\xi_k) / f'(x_k)$$

QUADRATIC CONVERGENCE

The next error is proportional to the *square* of the current one — correct digits roughly double per iteration. Requires M_k uniformly bounded; it fails when $f' \rightarrow 0$ near the root, the same phenomenon as an ill-conditioned root.

NEWTON AS FIXED POINT

$$g(x) = x - f(x)/f'(x), \quad x_{k+1} = g(x_k)$$
$$\text{quotient rule: } g'(x) = f(x)f''(x) / [f'(x)]^2$$
$$f(y) = 0 \Rightarrow g'(y) = 0$$

A second, independent proof of quadratic convergence via the fixed-point theory of block 06 — valid for *simple* roots, $f'(y) \neq 0$.

04Secant method

IDEA

Approximate the derivative in Newton's step by a difference quotient built from two points already computed.

$$f'(x_k) \approx (f(x_k) - f(x_{k-1})) / (x_k - x_{k-1})$$
$$x_{k+1} = x_k - f(x_k) \cdot (x_k - x_{k-1}) / (f(x_k) - f(x_{k-1}))$$

GEOMETRY

Newton uses the tangent at one point; the secant uses the line through $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$ — hence two starting points x_0, x_1 instead of one.

ADVANTAGE

No explicit computation of $f'(x)$ — valuable when the derivative is expensive, complicated, or not available in closed form.

COST

Superlinear but not quadratic convergence: order ≈ 1.618 , the golden ratio. Slower per iteration than Newton, but it never needs a reliable evaluation of f' .

05Newton for systems

PROBLEM

Given $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$, find $\tilde{x}$ with $f(\tilde{x}) = 0$: a vector of n functions vanishing simultaneously.

WHY NO BISECTION

The existence-of-zeros theorem does not generalise: in $\mathbb{R}^n$ there is no natural order relation, so " $a < x < b$ " is meaningless for vectors and there is no interval to halve.

TAYLOR IN $\mathbb{R}^n$ & ITERATION

$$f(z) = f(x) + J_f(x)(z - x) + O(\|z - x\|^2)$$
$$\text{set model to } 0: J_f(x_k) d = -f(x_k)$$
$$x_{k+1} = x_k + d$$

$J_f(x)$ is the Jacobian, the $n \times n$ matrix of partial derivatives generalising $f'(x)$. One cannot divide by it: a linear system is solved at each iteration.

STOPPING CRITERIA

Successive iterates: $\|x_{k+1} - x_k\| \leq C \cdot \text{Tol}$ with $C = \|x_k\|$. Residual: $\|f(x_k)\| \leq C \cdot \text{Tol}$ with $C = \|f(x_0)\|$ — C makes the test relative rather than absolute.

THEOREM — LOCAL CONVERGENCE

Let f be at least C^1 on a convex Ω , and assume: (1) $\exists x^* \in \Omega$ with $f(x^*) = 0$; (2) $J_f(x^*)$ invertible with $\|J_f^{-1}(x^*)\| < C$; (3) J_f Lipschitz-continuous near x^* . Then there is a ball $B(x^*, r)$ such that Newton converges for every $x_0 \in B(x^*, r)$. Purely local: no guarantee if you start far away.

LINK WITH THE LINEAR-SYSTEMS CHAPTER

Each Newton step is a linear solve (Gauss / LU, or iterative). To contain the cost the Jacobian is often frozen for several iterations instead of being recomputed and refactorised every time.

06Fixed-point iteration

DEFINITION

Rewrite $f(x) = 0$ as an equivalent $g(x) = x$. A value y with $g(y) = y$ is a fixed point of g . Continuity of g suffices for existence; differentiability is needed only to study convergence.

$$x_{k+1} = g(x_k)$$

ERROR ANALYSIS — LAGRANGE

$$e_k = x_k - y, \quad y = g(y)$$
$$e_{k+1} = g(x_k) - g(y) = g'(\xi_k) \cdot e_k$$
$$|e_{k+1}| = |g'(\xi_k)| \cdot |e_k|$$

Everything hinges on keeping that factor below 1.

THEOREM — LOCAL CONVERGENCE

g differentiable on $I_\eta = (y - \eta, y + \eta)$, y fixed point. If $|g'(x)| \leq M < 1$ for every x in an interval $I_\delta \subseteq I_\eta$, then $x_{k+1} = g(x_k) \rightarrow y$ for every $x_0 \in I_\delta$. Proof: iterate the estimate, $|e_k| \leq M^k |e_0| \rightarrow 0$. Local: $M < 1$ is needed only on I_δ , not on all of $\mathbb{R}$.

ASYMPTOTIC RATE

$$\lim_{k \rightarrow \infty} |e_{k+1}| / |e_k| = |g'(y)|$$

Linear convergence, faster the smaller $|g'(y)|$. If $|g'(y)| = 0$ exactly (as in Newton) the rate is better than linear — quadratic.

CONTRACTIONS

g is a **contraction** on $[a,b]$ if g is continuous, (1) $g: [a,b] \rightarrow [a,b]$ and (2) $|g(x_1) - g(x_2)| < |x_1 - x_2|$ for all $x_1, x_2 \in [a,b]$. **Contraction theorem:** then there exists a unique $y \in [a,b]$ with $g(y) = y$.

BUILDING g IN PRACTICE

$$g(x) = \lambda f(x) + x, \quad \lambda \neq 0 \Rightarrow (g(x) = x \Leftrightarrow f(x) = 0)$$
$$g'(x) = \lambda f'(x) + 1, \quad |g'(x)| < 1$$
$$\Rightarrow -2 < \lambda f'(x) < 0$$

λ must have the sign opposite to $f'(x)$, with $|\lambda| < 2/|f'(x)|$. If f' changes sign on the interval, no constant λ works everywhere and the interval must be restricted.

TWO DISTINCT THEOREMS · NEWTON AS SPECIAL CASE

The contraction theorem gives existence and uniqueness of the fixed point; the local theorem with $M < 1$ gives that the iteration actually reaches it from a close enough x_0 . With the variable choice $\lambda(x) = -1/f'(x)$ one recovers Newton, $g'(y) = 0$ — the locally optimal λ .

07Fixed point for systems

SAME STRUCTURE

With $f, g: \mathbb{R}^n \rightarrow \mathbb{R}^n$, transform $f(x) = 0$ into $g(x) = x$; the iteration is unchanged.

$$x_{k+1} = g(x_k)$$

CONVERGENCE CONDITION

$$\|J_g(x)\| < 1$$

The analogue of $|g'(x)| < 1$, written with the Jacobian of g — not of f — evaluated at the solution x .

ADDITIONAL REQUIREMENTS

The matrix norm must be compatible with the chosen vector norm — the same vector-norm / induced-norm pairing as in Chapter 1; J_g must be continuous; x_0 must lie in a sufficiently small neighbourhood of the solution.

METHOD	REQUIRES	ORDER	GUARANTEE	SYSTEMS
Bisection	f continuous, $f(a)f(b) < 0$	linear, rate $1/2$	always converges	no — no order in $\mathbb{R}^n$
Newton	f' , one x_0	quadratic	local only	yes — linear solve per step
Secant	two points, no f'	≈ 1.618 (golden ratio)	local only	quasi-Newton variants
Fixed point	g with $ g' \leq M < 1$	linear, rate $ g'(y) $	local; contraction $\Rightarrow$ unique root	yes — $\ J_g\ < 1$